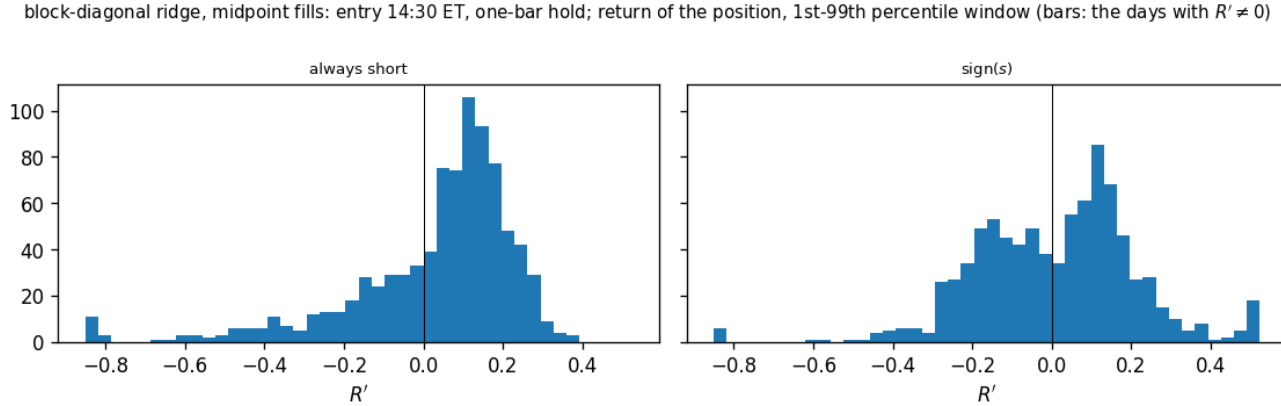


**ODTE nearest-OTM package: per-rule return summary, entry 14:30 ET, one-bar hold, models compared on the same 866 days**

|                                                                   | $n$ | mean  | std   | min   | max  | skew  | ex. kurt | $t$  | Sharpe <sub>ann</sub> | Sharpe <sub>ann</sub> <sup>x</sup> |
|-------------------------------------------------------------------|-----|-------|-------|-------|------|-------|----------|------|-----------------------|------------------------------------|
| <i>Short volatility (always short)</i>                            |     |       |       |       |      |       |          |      |                       |                                    |
| all models (no forecast)                                          | 866 | 0.019 | 0.300 | -5.15 | 0.38 | -7.56 | 108.09   | 1.86 | 1.00                  | -1.52                              |
| <i>sign(s) (<math>\pm 1</math> on the sign of <math>s</math>)</i> |     |       |       |       |      |       |          |      |                       |                                    |
| baseline (HAR + calendar OLS)                                     | 866 | 0.008 | 0.300 | -5.15 | 2.01 | -5.13 | 105.97   | 0.74 | 0.40                  | -2.15                              |
| block-diagonal ridge                                              | 866 | 0.009 | 0.300 | -5.15 | 2.01 | -5.43 | 106.17   | 0.88 | 0.48                  | -2.08                              |
| block-diagonal ridge, without the FOMC columns                    | 866 | 0.010 | 0.300 | -5.15 | 2.01 | -5.11 | 106.25   | 0.99 | 0.53                  | -2.02                              |
| LightGBM <sup>†</sup>                                             | 866 | 0.007 | 0.300 | -5.15 | 2.01 | -5.45 | 105.97   | 0.71 | 0.38                  | -2.17                              |
| XGBoost <sup>†</sup>                                              | 866 | 0.005 | 0.300 | -5.15 | 2.01 | -5.39 | 105.76   | 0.51 | 0.28                  | -2.28                              |
| lasso (causally tuned) <sup>†</sup>                               | 866 | 0.009 | 0.300 | -5.15 | 2.01 | -5.42 | 106.13   | 0.85 | 0.46                  | -2.09                              |
| lasso ( $\alpha = 10^{-4}$ )                                      | 866 | 0.007 | 0.300 | -5.15 | 2.01 | -5.41 | 105.93   | 0.67 | 0.36                  | -2.19                              |
| elastic net (causally tuned) <sup>†</sup>                         | 866 | 0.007 | 0.300 | -5.15 | 2.01 | -5.41 | 105.94   | 0.68 | 0.37                  | -2.19                              |



Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 14:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. One expiration day = one return; mid fill. Every  $t$  here is the plain  $t = \sqrt{n} \cdot \text{mean}/\text{std}$ , with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. Sharpe<sub>ann</sub> = (mean/std)  $\times \sqrt{252}$ ; every other column is daily. The one column this table adds to the deck’s is Sharpe<sub>ann</sub><sup>x</sup>, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed  $\alpha = 10^{-4}$ ; elastic net (causally tuned). <sup>†</sup> marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is  $s_t = \widehat{RV}_t - IV_{hr}^2 h_t w_t$ : the forecast against the implied variance of the same window. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining  $h_t$  hours — so every bar carries a signal and no bar sits flat.